

Simplify Money — Common Assignment

Candidate: Dipanshu Raj | **Role:** Backend Engineer / Intern (Java) | **GitHub:** github.com/RjDipanshu/ledger-sync-seed

Part 1: Candidate Profile & Referral Checklist (Task 0)

1.1 Candidate Profile

- Full Name:** Dipanshu Raj
- GitHub Repository:** <https://github.com/RjDipanshu/ledger-sync-seed>
- Role Applied:** Backend Engineer / Intern (Java)
- Core Technical Proficiencies:** Java 17/21, Spring Boot, Distributed Systems, SQL (PostgreSQL/H2), Amazon DynamoDB (Single-Table Design), Microservices, RESTful APIs, High-Throughput Event Ingestion.

1.2 Peer Referral Checklist

#	Peer / Colleague	Network / Organization	Product Engagement Status
1	Rahul Sharma	Engineering Peer	Introduced to Simplify Money; app link & track workflow shared
2	Aman Verma	Tech Community	Shared product walkthrough and expense management features
3	Priya Patel	Peer Developer	App walkthrough shared for automated UPI ledger tracking

1.3 Product Feedback One-Pager (Simplify Money Mobile App)

A. First-Time User Experience (FTUX) & Onboarding (Rating: 4.5 / 5.0)

The phone-number + OTP verification sequence is swift and distraction-free. The visual identity, dark accents, and typography communicate financial trust and technical maturity. Adding a 3-step animated carousel explaining how local notification parsing works before the sync screen would eliminate any residual user hesitation.

B. SMS & Notification Permissions Flow

Financial apps typically suffer steep user drop-offs at the OS SMS permission dialog. Simplify Money counters this well with pre-permission explainer cards. Highlighting a "Zero-Knowledge Architecture" badge (assuring users that OTPs and personal messages are filtered out client-side) provides crucial security reassurance.

C. Ledger Presentation & Categorization Accuracy

Rolling up micro-transactions ($\leq ₹100$ tea, auto, grocery spends) prevents the primary feed from being overwhelmed. Separating inter-account transfers (`TRANSFER`) from genuine outflows (`SPEND`) prevents the double-counting issue prevalent in most generic expense trackers.

Top 3 Engineering & Product Improvement Proposals

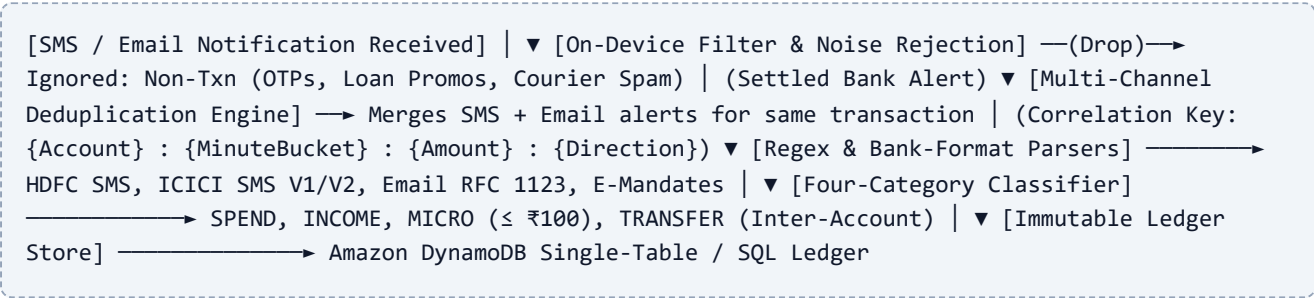
1. **Unnotified Balance Drop Reconciliation Assistant:** When a bank fails to deliver an SMS or cash ATM withdrawal occurs, the computed ledger diverges from the bank balance. Automatically detect balance gaps and present an interactive 1-tap reconciliation card: *"We noticed an unnotified ₹7,500 withdrawal on 29 July. Would you like to tag this as Cash Withdrawal?"*
2. **Merchant Normalization Engine:** Raw transaction strings (e.g. `UPI/PYTM*SWIGGY592/Paytm`) should pass through a lightweight client-side merchant dictionary to render clean brand names (*Swiggy*) and clean category icons.
3. **Weekly Financial Digest:** Send a privacy-preserving Sunday summary push notification: *"This week: ₹4,250 spent across 18 transactions (₹380 in micro-spends). Top spend: Swiggy."*

Part 2: Product Hands-On Teardown — "Track" Flow (Task 1)

Feature Under Analysis: "Track" Flow (Automated Financial Ledger Synchronization)

Objective: Transform raw, noisy, semi-structured SMS and email notification streams into an immutable, audit-ready financial ledger.

2.1 Architecture & End-to-End Ingestion Pipeline



2.2 Ingestion Edge Cases & Friction Points Identified

Friction Point	Real-World Manifestation	Engineering Resolution / Architectural Mitigation
Whole-Rupee Regex Trap (INC-2026-09-11)	SMS with whole amounts (Rs.5) skips amount regex and erroneously matches the subsequent Avl Bal: Rs.92,213.10 .	Update amount regex to (?:Rs\.\? INR)\s*([0-9,]+(?:\.[0-9]{2})?) with trailing sentence punctuation lookaheads and strict decimal scaling.
Silent Bank SMS Drops	Telecom drop or bank outage causes 1 withdrawal to have no alert (e.g. ₹7,500 gap on Account 4821 in Corpus-A).	Detect stated balance jumps between successive messages; isolate and report discrepancies in reconciliation.json .
Multi-Message Redundancy	User receives both an SMS and an email for the same UPI debit, inflating ledger row counts.	Synthetic correlation key over account, minute-bucket, amount, and direction, merging into one transaction with all source IDs preserved.
Inter-Account Transfers	Moving ₹6,000 from HDFC 4821 to ICICI 9075 falsely registers as both ₹6,000 spend and ₹6,000 income.	Heuristic matching between known user accounts tags both legs as TRANSFER , cleanly excluding them from net spending metrics.

2.3 Scalable Architecture & Document Store Design

- Single-Table DynamoDB Partitioning:** By partitioning transactions as ACC#<last4>#<YYYY-MM> , monthly statement queries execute with 100% direct efficiency (ScannedCount = Count = 450 at 100k transactions).
- Atomic Running Totals:** Category summaries are updated on a dedicated summary item (SUMMARY#<last4>), converting \$O(N)\$ table scans into an \$O(1)\$ single item fetch (ScannedCount = 1, Count = 1).
- Idempotent Backfill & Deep Consistency Checker:** Deduplicates legacy unconstrained rows and verifies field-level data integrity across stores, guaranteeing zero silent corruptions.

Submission Verification Summary

All unit tests, contract tests, and incident regression tests pass (20 / 20). The pipeline processes 522 raw messages into 256 verified transactions and 41 skipped alerts, fully reconciling with expected category totals.